Project: Worked examples
Title: Pipe pressure drop and pump duty
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

## 1. Scope and method

Friction and fitting losses through series pipe segments at a given flow, the static head, the system curve, the duty point against a pump curve entered as points, the affinity-law shift for a speed or impeller change, and the NPSH available at the pump suction.
Q = Φ/(ρ c_p ΔT) when a heat load is given; v = Q/(πD²/4); Re = ρvD/μ; f = 64/Re (Re < 2300) else Colebrook–White 1/√f = −2 log10(ε/3.7D + 2.51/(Re√f)) by fixed-point iteration (≤ 100 steps) with the Swamee–Jain explicit value as start and cross-check; h_f = f (L + L_eq)/D · v²/2g + K v²/2g; H_sys(Q) = Σdz + h_out + Σh_f(Q) with h_f ∝ Q² scaled from the design flow (closed circuit: Σdz = 0, the rise in the flow is recovered in the return); the water held in the pipes V_pipe = Σ π/4·D²·L; duty point where H_pump(Q) = H_sys(Q) (bisection); affinity: Q ∝ n D³, H ∝ n² D²; NPSH_a = p_atm/ρg + z_s − h_f,suction − p_v/ρg.
Ref: Crane Technical Paper 410 — Flow of fluids through valves, fittings and pipe
Ref: Çengel & Cimbala, Fluid Mechanics Ch. 8 — Internal flow: friction factor, minor losses, pump duty
Ref: CIBSE Guide C Section 4 — Flow of fluids in pipes and ducts (principles)
Ref: Kell, J. Chem. Eng. Data 20(1), 1975 — Density of liquid water

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| fluid | Fluid | water | | spec: SPEC-M-04 §3: chilled water, no glycol | any |
| T | Fluid temperature (mean; link B6 t_m for a heating loop) | 6 | degC | spec: SPEC-M-04 §3: 6 °C flow | 0 … 100 degC |
| Q | Design flow rate (blank when a heat load gives it) | 30 | m^3/h | spec: Schedule CHW-01 design flow | 0.01 … 100000 m^3/h |
| segments | Pipe segments in series | 3 row(s) | | drawing: DRG-M-110 rev C | any |
| segments[1] | | name = plantroom suction; L = 8 m; D = 100 mm; eps = 0.045 mm; K = 3; L_eq = 0 m; dz = 0 m; side = suction | | | |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| segments[2] | | name = riser DN80; L = 60 m; D = 80 mm; eps = 0.045 mm; K = 8; L_eq = 0 m; dz = 10 m; side = discharge | | | |
| segments[3] | | name = index run DN65; L = 45 m; D = 65 mm; eps = 0.045 mm; K = 12; L_eq = 6 m; dz = 2 m; side = discharge | | | |
| circuit | Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored) | open (the pump lifts the water by the rises entered) | | default | any |
| h_out | Pressure head required at the outlet | 3 | m | spec: Terminal unit: 30 kPa ≈ 3 m | 0 … 1000 m |
| v_max | Velocity limit (warning) | 2.5 | m/s | spec: CIBSE Guide C practice limit, assumed | 0.5 … 10 m/s |
| Q_pump | Pump curve: flow points | [0, 20, 40, 60] | m^3/h | spec: Pump datasheet P-01 rev 2, curve points | any |
| H_pump | Pump curve: head points | [40, 36, 24, 4] | m | spec: Pump datasheet P-01 rev 2, curve points | any |
| n_ratio | Speed ratio $n_2/n_1$ (affinity) | 0.9 | | assumption: Inverter turn-down to 90 % speed. | 0.2 … 2 |
| D_ratio | Impeller diameter ratio $D_2/D_1$ (affinity) | 1 | | assumption: Impeller unchanged. | 0.7 … 1.1 |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| eta_pump | Pump efficiency at duty (for shaft power) | 0.68 | | spec: Pump datasheet P-01 rev 2 | 0.2 … 0.95 |
| p_atm | Atmospheric pressure (NPSH) | 101.325 | kPa | assumption: Sea level. | 50 … 110 kPa |
| z_s | Static head at the pump suction (+ flooded, − lift) | 2 | m | drawing: DRG-M-110 rev C | -10 … 100 m |
| NPSH_r | NPSH required by the pump (optional) | 3 | m | spec: Pump datasheet P-01 rev 2 | 0 … 50 m |

## 3. Assumptions

A1. Steady, incompressible, fully developed flow; all segments carry the full design flow in series. (Darcy–Weisbach.)
A2. Fitting losses scale with velocity head (K method) or add pipe length (equivalent-length method); both may be used on one segment. (Crane TP-410 practice.)
A3. The system curve scales friction losses with $Q^2$ from the design flow (turbulent assumption); the static part is independent of flow. ($H_{sys} = H_{static} + k\,Q^2$.)
A4. Water properties are from published correlations at atmospheric pressure (0–100 °C); any other fluid is entered with its own density and viscosity and their provenance. (Kell 1975; Vogel fit; Hyland & Wexler 1983.)
A5. n_ratio (Speed ratio $n_2/n_1$ (affinity)) is taken as 0.9. (Inverter turn-down to 90 % speed.)
A6. D_ratio (Impeller diameter ratio $D_2/D_1$ (affinity)) is taken as 1. (Impeller unchanged.)
A7. p_atm (Atmospheric pressure (NPSH)) is taken as 101.325 kPa. (Sea level.)

## 4. Working

[3] rho = Kell (1975) density of water at T — Kell 1975
= T = 6 °C
= 999.94 kg/m³
Density

[4] mu = Vogel fit □ = 0.02939 exp(507.88/(T_K − 149.3)) mPa·s
= T_K = 279.15 K
= 1.4684 mPa·s
Dynamic viscosity

[5] p_v = Hyland–Wexler saturation pressure
= T = 6 °C
= 0.93525 kPa
Vapour pressure (NPSH)

[9] Q_design = Q
= 30 m³/h
= 30 m³/h
Design flow (as entered)

[10] v_s1 = plantroom suction: Q / (π D²/4)
= 30 m³/h / (π × 100² mm² / 4)
= 1.061 m/s
Velocity, plantroom suction

[11] Re_s1 = plantroom suction: □ v D / □
= 999.94 × 1.061 × 0.1 / 0.0014684
= 72253
Reynolds number, plantroom suction

[12] f_s1 = plantroom suction: Darcy friction factor
= Colebrook-White, 10 iterations (Swamee-Jain start 0.021152)
= 0.021093
Friction factor, plantroom suction
[13] f_sj_s1 = plantroom suction: Swamee-Jain check
= 0.25 / log10(/3.7D + 5.74/Re^0.9)²
= 0.021152
Explicit cross-check, plantroom suction
[14] h_f_s1 = plantroom suction: f (L + L_eq)/D · v²/2g + K v²/2g
= 0.021093 × (8 + 0) / 0.1 × 0.057399 + 3 × 0.057399
= 0.26906 m
Head loss, plantroom suction
[15] v_s2 = riser DN80: Q / (π D²/4)
= 30 m³/h / (π × 80² mm² / 4)
= 1.6579 m/s
Velocity, riser DN80
[16] Re_s2 = riser DN80:  v D / 
= 999.94 × 1.6579 × 0.08 / 0.0014684
= 90316
Reynolds number, riser DN80
[17] f_s2 = riser DN80: Darcy friction factor
= Colebrook-White, 10 iterations (Swamee-Jain start 0.020946)
= 0.020849
Friction factor, riser DN80
[18] f_sj_s2 = riser DN80: Swamee-Jain check
= 0.25 / log10(/3.7D + 5.74/Re^0.9)²
= 0.020946
Explicit cross-check, riser DN80
[19] h_f_s2 = riser DN80: f (L + L_eq)/D · v²/2g + K v²/2g
= 0.020849 × (60 + 0) / 0.08 × 0.14014 + 8 × 0.14014
= 3.3123 m
Head loss, riser DN80
[20] v_s3 = index run DN65: Q / (π D²/4)
= 30 m³/h / (π × 65² mm² / 4)
= 2.5113 m/s
Velocity, index run DN65
[21] Re_s3 = index run DN65:  v D / 
= 999.94 × 2.5113 × 0.065 / 0.0014684
= 111160
Reynolds number, index run DN65
[22] f_s3 = index run DN65: Darcy friction factor
= Colebrook-White, 9 iterations (Swamee-Jain start 0.020964)
= 0.020834
Friction factor, index run DN65
[23] f_sj_s3 = index run DN65: Swamee-Jain check
= 0.25 / log10(/3.7D + 5.74/Re^0.9)²
= 0.020964
Explicit cross-check, index run DN65
[24] h_f_s3 = index run DN65: f (L + L_eq)/D · v²/2g + K v²/2g
= 0.020834 × (45 + 6) / 0.065 × 0.32155 + 12 × 0.32155
= 9.1151 m
Head loss, index run DN65

[25] h_f_total = Σ h_f
= 3 segment(s)
= 12.696 m
Total friction and fitting loss at the design flow
[27] V_pipe = Σ π/4·D²·L
= 3 segment(s)
= 513.75 L
Water held in the pipe segments
[28] H_static = Σ dz + h_out
= 12 + 3
= 15 m
Static head
[29] H_sys = H_static + h_f_total
= 15 m + 12.696 m
= 27.696 m
System head at the design flow
[30] dp_total = rho * 9.80665 m/s^2 * h_f_total
= 999.94 kg/m³ × 9.80665 m/s^2 × 12.696 m
= 124.5 kPa
Friction pressure drop at the design flow
[31] dp_sys = rho * 9.80665 m/s^2 * H_sys
= 999.94 kg/m³ × 9.80665 m/s^2 × 27.696 m
= 271.59 kPa
System pressure at the design flow
[32] P_hyd = rho * 9.80665 m/s^2 * Q_design * H_sys
= 999.94 kg/m³ × 9.80665 m/s^2 × 30 m³/h × 27.696 m
= 2.2633 kW
Hydraulic power at the design point
[38] Q_duty = duty flow (as entered)
= H_pump(Q) = H_sys(Q), bisection
= 31.569 m³/h
Duty flow as entered
[39] H_duty = duty head (as entered)
= H_sys(31.569 m³/h)
= 29.059 m
Duty head as entered
[40] P_shaft = Σ g Q H / η
= 999.94 × 9.8066 × 0.008769 × 29.059 / 0.68
= 3.6747 kW
Shaft power at duty as entered
[41] affinity = $Q_2 = Q_1 (n_2/n_1)(D_2/D_1)^3$, $H_2 = H_1 (n_2/n_1)^2(D_2/D_1)^2$
= $n_2/n_1$ = 0.9, $D_2/D_1$ = 1
= "curve scaled"
Affinity laws applied to the pump curve
[42] Q_duty2 = duty flow (after the affinity shift)
= H_pump(Q) = H_sys(Q), bisection
= 26.238 m³/h
Duty flow after the affinity shift
[43] H_duty2 = duty head (after the affinity shift)
= H_sys(26.238 m³/h)
= 24.712 m
Duty head after the affinity shift

[44] P_shaft2 = □ g Q H / □
= 999.94 × 9.8066 × 0.0072883 × 24.712 / 0.68
= 2.5972 kW
Shaft power at duty after the affinity shift
[47] h_f_suction = □ h_f on the suction side
= 1 suction segment(s)
= 0.26906 m
Suction-side friction loss
[48] NPSH_a = p_atm / (rho * 9.80665 m/s^2) + z_s - h_f_suction - p_v / (rho * 9.80665 m/s^2)
= 101.32 kPa / (999.94 kg/m³ × 9.80665 m/s^2) + 2 m − 0.26906 m − 0.93525 kPa / (999.94 kg/m³ × 9.80665 m/s^2)
= 11.968 m
NPSH available
[50] u_npsh = NPSH_r / NPSH_a
= 3 m / 11.968 m
= 0.25066
NPSH utilisation (required / available)
[51] v_peak = highest segment velocity
= 2.5113 m/s
= 2.5113 m/s
Highest velocity
[52] u_v = v_peak / v_max
= 2.5113 / 2.5
= 1.0045
Velocity utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| v_s1 | 1.061 m/s | Velocity, plantroom suction |
| Re_s1 | 72253 | Re, plantroom suction |
| f_s1 | 0.021093 | f, plantroom suction |
| f_sj_s1 | 0.021152 | f (Swamee–Jain), plantroom suction |
| h_f_s1 | 0.26906 m | h_f, plantroom suction |
| v_s2 | 1.6579 m/s | Velocity, riser DN80 |
| Re_s2 | 90316 | Re, riser DN80 |
| f_s2 | 0.020849 | f, riser DN80 |
| f_sj_s2 | 0.020946 | f (Swamee–Jain), riser DN80 |
| h_f_s2 | 3.3123 m | h_f, riser DN80 |
| v_s3 | 2.5113 m/s | Velocity, index run DN65 |
| Re_s3 | 111160 | Re, index run DN65 |
| f_s3 | 0.020834 | f, index run DN65 |
| f_sj_s3 | 0.020964 | f (Swamee–Jain), index run DN65 |
| h_f_s3 | 9.1151 m | h_f, index run DN65 |
| h_f_total | 12.696 m | Σ h_f |
| H_static | 15 m | Static head |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| V_pipe | 513.75 L | Water held in the pipes |
| H_sys | 27.696 m | System head at Q |
| dp_total | 124.5 kPa | Friction Δp |
| dp_sys | 271.59 kPa | System Δp |
| Q_design | 30 m³/h | Design flow |
| P_hyd | 2.2633 kW | Hydraulic power |
| Q_duty | 31.569 m³/h | Duty flow as entered |
| H_duty | 29.059 m | Duty head as entered |
| P_shaft | 3.6747 kW | Shaft power as entered |
| Q_duty2 | 26.238 m³/h | Duty flow after the affinity shift |
| H_duty2 | 24.712 m | Duty head after the affinity shift |
| P_shaft2 | 2.5972 kW | Shaft power after the affinity shift |
| NPSH_a | 11.968 m | NPSH available |
| v_peak | 2.5113 m/s | Highest velocity |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| FAIL | Velocity ≤ limit | u_v = 1.0045 | u_v ≤ 1 | 1 | |
| PASS | NPSH_r ≤ NPSH_a | u_npsh = 0.25066 | u_npsh ≤ 1 | 0.251 | |
| PASS | Duty point exists (as entered) | 31.569 m³/h | 29.059 m | 0 | |
| PASS | Duty point exists (after the affinity shift) | 26.238 m³/h | 24.712 m | 0 | |

## 7. Applicability

WARNING — input.circuit: circuit (Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored)) was not entered; the template default 'open (the pump lifts the water by the rises entered)' is used.
WARNING — velocity: segments: index run DN65 runs at 2.5113 m/s, above the 2.5 m/s limit.

## 8. Verification

Template bs.pipe-pump@1.3.0 Content hash d3c3c38f4c81cf2d Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0